

Lesson 11 — Build the Model & If / Then

Matches LEGO Teacher Portal lesson: "Game On".

Objective (I can...): I can build the conditionals model and use an if/then so the machine decides whether to act. **CS concept:** Selection; boolean conditions in an if/then · **Builds on:** Lesson 10 (loops set the stage for decisions inside them).

AI Connection (2 min): A classifier's output drives a decision: "IF confidence that it's a dog > 80%, THEN label it dog." *First-principles question:* "What should the program do when confidence is right at the borderline?"

Materials: LEGO CS & AI kit + device per pair, trigger objects / color cards, notebooks. **This is the unit's build lesson** — build the Conditionals model now and keep it assembled through Lesson 15.

60-minute flow - Warm-up (8): "True statement, now act" — coach reads conditions; pairs state whether each is true right now and what action should follow. Introduce *boolean*: a condition is either true or false. - **Concept + demo (10):** Demo an if/then: if the distance sensor reads under a value, then stop; otherwise nothing. Show the live sensor reading so the condition's truth is visible. - **Build/Code (30):** Pairs **build the Conditionals model** (the unit's one build), then program one if/then driven by a sensor (object near → act). They write the condition in words in the notebook. *Swap at 15 min.* - **Challenge (7):** Change the condition's threshold and verify the action fires only when the condition is true. - **Share + pack-up (5):** Pairs demo the true case and the "nothing happens" case and read their condition aloud. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator reads the condition and watches the live value. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — a palette card for the if/then block and comparison operators. *Extension / fast-finisher* — make the "then" a two-step action. *Reteach* — move the object while narrating "true... false..." against the live reading.

Exit check: Pair's model acts only when its boolean condition is true. **At-home (optional):** Write three if/then rules that govern a device in your home; state the condition precisely. ~10 min.
